GATI VARSHNEY

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Education

JSS Academy of Technical Education, Noida, India — *2023 – 2027*
Bachelor of Technology in Computer Science and Engineering (Data Science) — ***GPA: 8.7/10***

Experience

Software Engineer Intern, Quin Que Inc. (Japan, Remote) — *Sep 2026*

- Shipped 6 admin screens for Rapilogi, a multi-tenant warehouse platform, across the OpenAPI contract, PostgreSQL migrations, Go handlers and React, with MFA login and Row-Level Security for tenant isolation.
- Caught a tenant-provisioning defect in the design documents before writing code, and generalized the fix into a reusable tenant-preset field instead of a one-off patch.
- Fixed SSE reconnection and React Native crashes outside assigned scope; 4 PRs merged through code review.

Google Summer of Code 2026 Contributor, The Linux Foundation — *May 2026 – Aug 2026*
AI-Driven Printer Compatibility & Recommendation Portal **[Final Report]**

- Built a recommendation engine over 6,657 printers that loads 6 KB per page instead of 24 MB by ranking them on 415 features with IDF-weighted cosine similarity and sharding output per printer.
- Diagnosed that 86.8% of similarity scores saturated at 1.0, making rankings meaningless; cut it to 0% with evidence damping and capability-conflict penalties, guarded by a harness that fails on any metric drift.
- Engineered a natural-language printer assistant that runs deterministically in-browser at zero page-load cost and flags data gaps instead of guessing; verified by 384 tests including a 187-utterance corpus.

Open Source Developer, OpenPrinting **[Merged PRs]** — *Nov 2025 – Aug 2026*

- Delivered production site search across 200+ posts with no search server by compiling posts into a static JSON index via a build-time AST pipeline (unified, remark-parse), queried in-browser by MiniSearch with fuzzy matching.
- Migrated 200+ Jekyll posts and 118 media files to Next.js with legacy-URL redirects, and shipped the author system and homepage; 14 PRs merged through maintainer review.
- Fixed production 404s across 7.5K+ routes, including RSS links consumed by LWN, by tracing them to a deploy-time config overriding the repository's, and added a CI check blocking deploys with broken internal links.

Projects

Contributable | Next.js, TypeScript, GraphQL, GitHub Actions **[Live]** **[GitHub]** — *2026*

- Launched a live platform (2K+ visitors) ranking 1,500+ repositories from Google Summer of Code organizations by newcomer-friendliness: first-reply time, outside-contributor merge rate, starter issues.
- Served fresh data with no database and no GitHub call on any page view by scheduling a GitHub Actions sweep of the GraphQL API that fetches only changes and commits snapshots to a data branch.
- Made every published figure defensible by modeling reply and merge times with Kaplan-Meier survival estimates, filtering bots, withholding any figure under 5 samples and linking each number to its source.

cidx | Python, tree-sitter, SQLite, MCP **[Live]** **[GitHub]** — *2026*

- Published a zero-config code index for AI coding agents to PyPI, validated on Django (3,000+ files), by parsing Python and TypeScript/JavaScript with tree-sitter into a SQLite symbol index served over MCP.
- Guaranteed incremental updates match a cold rebuild exactly by enforcing content-hash invalidation and per-file SQLite transactions, verified by Hypothesis property tests and crash injection in CI on 3 operating systems.
- Exposed five read-only MCP tools over stdio with token-budgeted responses, freshness stamps and a grep fallback on every miss, so agents get locations and signatures instead of whole files.

Achievements

- Winter of Code 5.0: Ranked Top 20 of 2,800+ participants
- OpenPrinting Hall of Fame: Recognized by maintainers for contributions to the project website **[Listing]**
- LeetCode Knight: Contest rating 1925, top 4% of contest participants globally **[Profile]**

Technical Skills

Languages: C++, C, Python, TypeScript, JavaScript, SQL
Core Subjects: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Backend: Node.js, Express.js, REST APIs, GraphQL, OpenAPI, PostgreSQL, MongoDB, SQLite, JWT & NextAuth
Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3
AI Integration: Google Gemini API, Model Context Protocol (MCP), schema-constrained structured outputs (Zod)
Developer Tools: Git, GitHub Actions (CI/CD), Linux, tree-sitter, pytest, Hypothesis, Puppeteer